

Abigail Hepburn

abigailzhepburn@gmail.com | linkedin.com/in/abigail-hepburn | abbyhepburn.github.io | github.com/abbyhepburn/

Education

University of Florida, College of Engineering

Expected May 2029

Bachelor of Science in Computer Engineering - Minor in Sales Engineering

GPA: 3.87

Relevant Coursework: Programming Fundamentals 1& 2 (Python & C++), Discrete Structures

Current Coursework: Data Structures & Algorithms (C++), Signals & Systems (MATLAB)

Experience

Data Engineer Intern

August 2026-January 2027

BeachLens

- Collected and structured data from various beaches and their local businesses to support features for the BeachLens app, which currently has more than 20,000 downloads.
- Created a script to calculate the geolocation, noaa station & layer id of any beach using Python's geocopy library, which allowed 30+ beaches from 3 different states to be onboarded faster.
- Built automated data pipelines using Python, Jina, and the Navigator API to extract website data, format it, and store it directly in Supabase. Created a faster way to collect the public information of businesses local to certain beaches.

Power Delivery Engineer Intern

NextEra Energy

May 2026-August 2026

- Cleaned, maintained, and validated the distribution operations database for a grid serving over 6 million customer accounts to improve data integrity, system accuracy, and analysis reliability.
- Designed data-driven solutions for electrical grid infrastructure by interpreting load information, mapping circuit pathways, and technical schematics; supporting 30+ distribution projects overall.
- Audited technical asset specifications and evaluated power delivery components to ensure all designs were fully constructed in the field, eliminating operational risk and rework costs.

Research Assistant

UF Data Studio

October 2025-May 2026

- Conducted research on LLM-based synthetic prediction generation using Python, producing structured datasets stored in CSV format.
 - Designed and evaluated prompting strategies (few-shot, zero-shot, chain-of-thought) to generate multi-domain prediction data (finance, health, policy, weather, sports).
 - Executed Python data pipelines with the NaviGator Toolkit to generate 10-15 batches of data per week.
-

Projects

Face Detection System | Python, OpenCV, MediaPipe, NumPy, PySide6

April 2026

- Created a real-time computer vision app using OpenCV and MediaPipe for facial landmark mapping and NumPy for high-performance image array processing.
- Integrated OpenCV to manage live video capture and image processing, used MediaPipe to extract 3D facial landmarks, NumPy for optimized multi-dimensional array manipulation, and PySide6 to display the interface

FlexRight | Python, MongoDB, Ultralytics, OpenCV, Gradio, CSS

January 2026

- Hackathon Team Project, developed an AI-driven fitness tool to provide real-time form correction.
- Engineered complex backend logic to calculate joint angles and track skeletal positioning from the live coordinates given by YOLO, bridging high-performance pose estimation with a secure MongoDB data logging system.
- Specialized in managing our database and integrating OpenCV's image processing pipelines to overlay dynamic visual cues and skeletal maps onto the user's video feed.

Support Bot | OpenAI, Python, Gradio

October 2025

- Designed and built a document-aware AI chatbot that utilizes a RAG pipeline to answer the user's questions based on the uploaded documents.
 - Allows for the user to upload any .pdf, .docx, or .txt document, extracts the file's text, and checks for similarities.
-

Leadership

Technical Development Chair - Software Lead | National Society of Black Engineers

- Conducting software-focused technical workshops for a club of over 200+ members
- Co-leading the NSBE design team of 15+ members by planning and guiding the software side of developing a self-balancing robot

Software Design Team Lead | Real World Engineering

- Worked with a team of Mechanical & Electrical Engineers to create a robotic hand with 4-degrees of freedom
 - Key tasks included developing control logic using Python functions to implement forward and inverse kinematics
-

Skills

Languages: Python, C++, JavaScript, HTML, CSS, SQL (MySQL, PostgreSQL), Swift, MATLAB

Developer Tools: PyCharm, VS Code, GitHub, Git, Clion

Certifications: Java, Onshape CAD Fundamentals, Device Configuration, HTML/CSS/Javascript